Evaluation Form (1)

Title: Evaluation of the Proposed LLM-Based Agentic Architectural Framework for Smart Learning Environments.
Focus of the Interview: Learning Design Agent
Date:

We are conducting a short-term study to evaluate our proposed Large Language Model (LLM)-based agentic architectural framework that integrates three LLM-based agents (as shown in Figure 1) guided by SOLO (Structure of Observed Learning Outcome) taxonomy and Knowledge Components (KCs) into POLARIS which is a Smart Learning Environment (SLE). The framework is designed to strengthen the link between your learning design and learning analytics, capture the depth of student understanding through the SOLO taxonomy, and generate personalized, context-aware reactions for learners. This interview focuses on the *first LLM-based agent*, called the **Learning Design Agent**. Its purpose is to connect the teacher's learning design intentions with the analytics and reaction layers of the POLARIS. It makes sure that learning materials are turned into structured and pedagogically meaningful KCs. The agent analyzes teacher-provided materials, identifies the KCs, annotates them with mastery examples aligned to the SOLO taxonomy, defining the depth of reasoning expected, and map the KCs with learning activities.

Here is a short explanation of key concepts:

- ***LLMs**:* LLMs are advanced type of artificial intelligence trained on text data to understand and generate human-like language.
- ***KCs**:* These are small pieces of knowledge or skills that students are expected to learn. For example, in Organization of school space, a KC could be "*Strategies for zoning in inclusive playgrounds*". A learning objective describes what students should achieve by the end of the course, such as explaining, analyzing, or designing something. In contrast, a KC represents a specific piece of knowledge or skill that students must acquire in order to successfully perform learning tasks. In simple terms, learning objectives define the goals of learning, while KCs are the knowledges or skills students need to master to reach those goals. They help connect teachers' intentions with how student learning is analyzed and tracked in learning systems.
- ***SOLO taxonomy**:* This is a framework to describe the depth of student understanding. It has five levels:
    1. Pre-structural – no real understanding.
    2. Uni-structural – focuses on just one detail.
    3. Multi-structural – knows several details but doesn't connect them.
    4. Relational – connects details into a meaningful whole.
    5. Extended Abstract – goes beyond, generalizes, or creates new ideas.

Learning Design Agent

Step 1:

Please rate the 12 generated KCs and their corresponding SOLO-aligned mastery examples on a scale from 1 (Strongly Disagree) to 6 (Strongly Agree).

KC001: ...

Question	**Strongly Disagree (1)**	**Disagree (2)**	**Slightly Disagree (3)**	**Slightly Agree (4)**	**Agree (5)**	**Strongly Agree (6)**
The **`KC001`** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **`KC001`** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **`KC001`**.						

KC002: ...

Question	**Strongly Disagree (1)**	**Disagree (2)**	**Slightly Disagree (3)**	**Slightly Agree (4)**	**Agree (5)**	**Strongly Agree (6)**
The **`KC002`** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **`KC002`** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **`KC002`**.						

KC003: ...

Question	**Strongly Disagree (1)**	**Disagree (2)**	**Slightly Disagree (3)**	**Slightly Agree (4)**	**Agree (5)**	**Strongly Agree (6)**
The **`KC003`** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **`KC003`** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **`KC003`**.						

KC004: ...

Question	**Strongly Disagree (1)**	**Disagree (2)**	**Slightly Disagree (3)**	**Slightly Agree (4)**	**Agree (5)**	**Strongly Agree (6)**
The **`KC004`** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **`KC004`** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **`KC004`**.						

KC005: ...

Question	**Strongly Disagree (1)**	**Disagree (2)**	**Slightly Disagree (3)**	**Slightly Agree (4)**	**Agree (5)**	**Strongly Agree (6)**
The **`KC005`** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						

The **KC005** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **KC005**.						

KC006: ...

Question	**Strongly Disagree (1)**	**Disagree (2)**	**Slightly Disagree (3)**	**Slightly Agree (4)**	**Agree (5)**	**Strongly Agree (6)**
The **KC006** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **KC006** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **KC006**.						

KC007: ...

Question	**Strongly Disagree (1)**	**Disagree (2)**	**Slightly Disagree (3)**	**Slightly Agree (4)**	**Agree (5)**	**Strongly Agree (6)**
The **KC007** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **KC007** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **KC007**.						

KC008: ...

Question	**Strongly Disagree (1)**	**Disagree (2)**	**Slightly Disagree (3)**	**Slightly Agree (4)**	**Agree (5)**	**Strongly Agree (6)**
The **KC008** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **KC008** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **KC008**.						

KC009: ...

Question	**Strongly Disagree (1)**	**Disagree (2)**	**Slightly Disagree (3)**	**Slightly Agree (4)**	**Agree (5)**	**Strongly Agree (6)**
The **KC009** proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The **KC009** proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for **KC009**.						

KC010: …

Question	Strongly Disagree (1)	Disagree (2)	Slightly Disagree (3)	Slightly Agree (4)	Agree (5)	Strongly Agree (6)
The `KC010` proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The `KC010` proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for `KC010`.						

KC011: …

Question	Strongly Disagree (1)	Disagree (2)	Slightly Disagree (3)	Slightly Agree (4)	Agree (5)	Strongly Agree (6)
The `KC011` proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The `KC011` proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for `KC011`.						

KC012: …

Question	Strongly Disagree (1)	Disagree (2)	Slightly Disagree (3)	Slightly Agree (4)	Agree (5)	Strongly Agree (6)
The `KC012` proposed by the agent align with my intended learning outcomes (it is pedagogically meaningful).						
The `KC012` proposed by the agent is clear.						
The SOLO-aligned mastery examples (uni-structural, multi-structural, relational, etc.) are appropriate for `KC012`.						

Step 2:

Now is time to provide us with the mapping of the 12 KCs to the learning activities with IDs.

- **KC001** – … → **Activity ID:**
- **KC002** – … → **Activity ID:**
- **KC003** – … → **Activity ID:**
- **KC004** – … → **Activity ID:**
- **KC005** – … → **Activity ID:**
- **KC006** – … → **Activity ID:**
- **KC007** – … → **Activity ID:**
- **KC008** – … → **Activity ID:**
- **KC009** – … → **Activity ID:**
- **KC010** – … → **Activity ID:**
- **KC011** – … → **Activity ID:**
- **KC012** – … → **Activity ID:**